

Do adolescents who handwrite as their normal way of working show increased connectivity in the parietal region compared to adolescents who type as their normal way of working?

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Abstract

- HG-EEG study comparing alpha and theta oscillations in the parietal brain regions of adolescents when typing or handwriting in response to an audio stimulus.
- Results will inform and influence UK educational practices to ensure school use of optimal learning modalities.

Introduction

- Optimal learning is associated with decreases in alpha oscillations and simultaneous increases in theta oscillations in the parietal cortex (Riddle et al., 2020).
- Such activity is present in the postcentral gyrus and posterior parietal regions when handwriting but absent when typing (Marano et al., 2025).



- Consequences of current trends of increased typing by adolescents in schools unclear as previous research:
 - Did not study adolescents → Gap in understanding
 - Ignored pre-existing typing skills → Confounding variable
 - Required participants to type with one finger → Lab effect
- (Askvik et al., 2020; van der Weel & van der Meer, 2024)

The proposed research will aim to demonstrate handwriting, not typing, optimises adolescent brains for learning by:

- Focusing on adolescents.
- Comparing fluent handwriting with fluent typing.
- Considering effects of typing as a normal way of working.

Research Design & Method

- Participants = right-handed, male and female 15-year-old UK school pupils.
- N=63 (based upon *a priori* analysis in G*Power 3.1) in three equal groups: typists typing; hand-writers handwriting; hand-writers typing.

Procedure:



Ethics:

- Safety, comfort, wellbeing and consent of adolescent participants prioritised.
- BPS Guidance followed throughout e.g. information sheets/consent forms.
- Parent/carers can observe at all times and researchers fully DBS checked

Discussion & Limitations

Decreased alpha oscillations and simultaneous increases in theta oscillations in handwriting group only will demonstrate handwriting, not typing (regardless of habituation), optimises adolescent learning, influencing educational policy.

- Limitations include the potential for cross-hemispheric noise on the EEG due to typists using both hands but hand-writers predominantly using one.

References

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